



Alternatives, Evaluation and the Master Facility Plan.

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Developing alternatives

Design = *CREATIVE* part of the process
(can feel FUZZY)

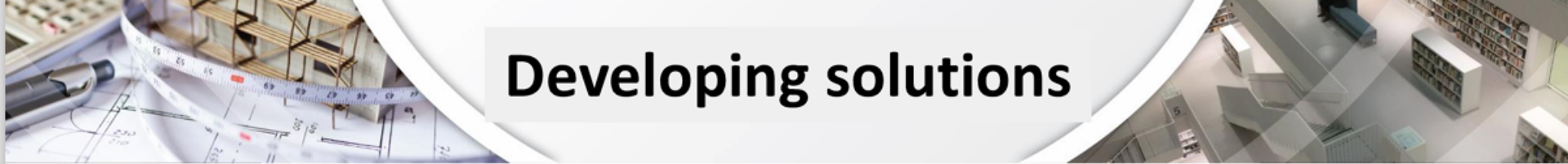
- May feel like going back and forth (zoom in zoom out)
– YES that is OK!
- You can detail only certain areas/aspects
- Your base data/information must be reliable
- Some procedures may help e.g. SLP (Munther)
- There is iteration between steps, even going back to recalculate, measure, fit, move, redesign process
- To test feasibility – SIMULATION models may help

Developing alternatives

- Involve people
- Think outside of box
- *Brainstorm* and *Bounce* ideas
- Don't think solution too early
- Don't fail to see the forest for the trees
 - Don't *judge too quickly*

The DESIGN process ask for a SELECTION to be made, it does not ask for the BEST design! Note that.

Goal therefore is to 'satisfy' not 'optimize'

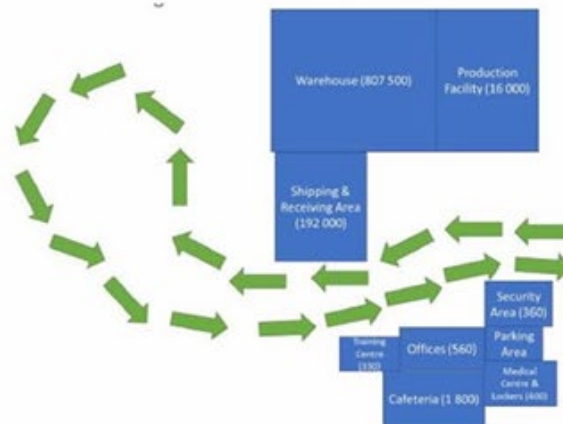
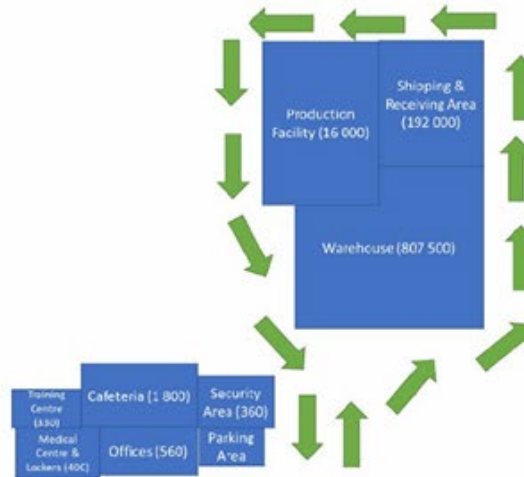


Developing solutions

- This is the phase in the Facility Design process, where you really need your creative thinking skills!
- It is important to understand that for facilities, there is no one, right answer.
- As a result, the more time you spend on the creative, brainstorming processes around developing alternative layouts, the more likely you are to come up with a successful design.

Key points of this chapter have been highlighted in these slides but to get the real essence of developing layout alternatives, I would strongly recommend reading through the chapter at least once.

Block layouts



See how useful these can be when comparing different layout alternatives. We usually do this before we develop detail internal layouts.

Evaluating alternatives *(Ch 11)*

- Remember at start...*design requirements?*
- Now is the time to **measure against those** – WHY?
 - so that Evaluation is not a **Guess....**
 - but a Decision based on **Facts**
 - and **Systematically** done
- Someone else must be able to also see why and how decisions were made (this is tricky if you are biased)
 - Involve variety of people to help judge
 - Don't be afraid to combine ideas also
 - If alternative ideas are not included in the comparisons, then it cannot (and will not) be selected!
 - Try and focus on data driven solutions and not management ego's or feelings

Evaluation tools/techniques

- Start with Requirements e.g. was it to...
 - Minimize distance traveled...**so did it?** show it
 - Meet demand.....ditto
 - Maximize space utilization.....ditto
 - Increase machine utilization.....ditto
 - Decrease material handling costs.....ditto
 - Decrease material movement time....ditto
- Pros/Cons or Advantages/Disadvantages (Fig 11.1&2)
- Ranking (Fig 11.3 illustrates example)
- Weighted factor comparison (Fig 11.4 and Ex 11.1)
(quantifying subjective factors are tricky)
- (or if you wish Analytical Hierarch Process AHP)
- Economic evaluations

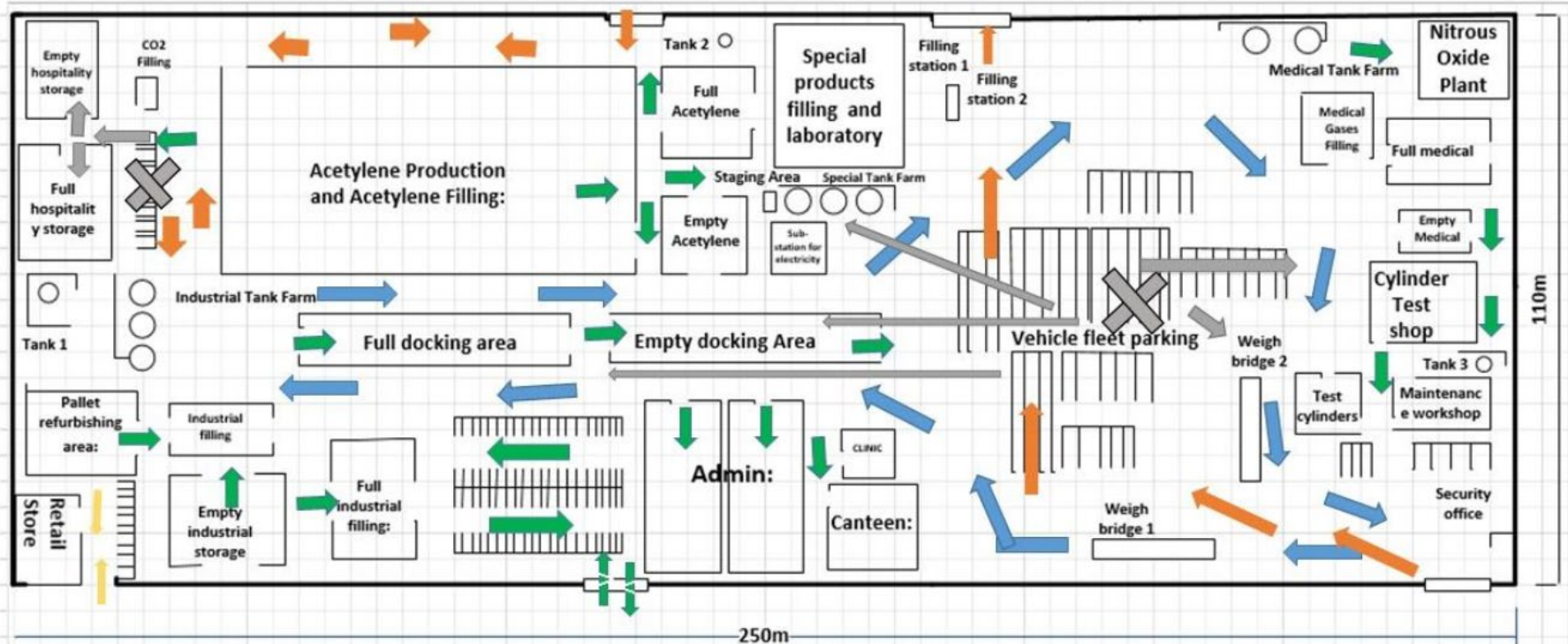
The Master Facility Layout *(Ch 12)*

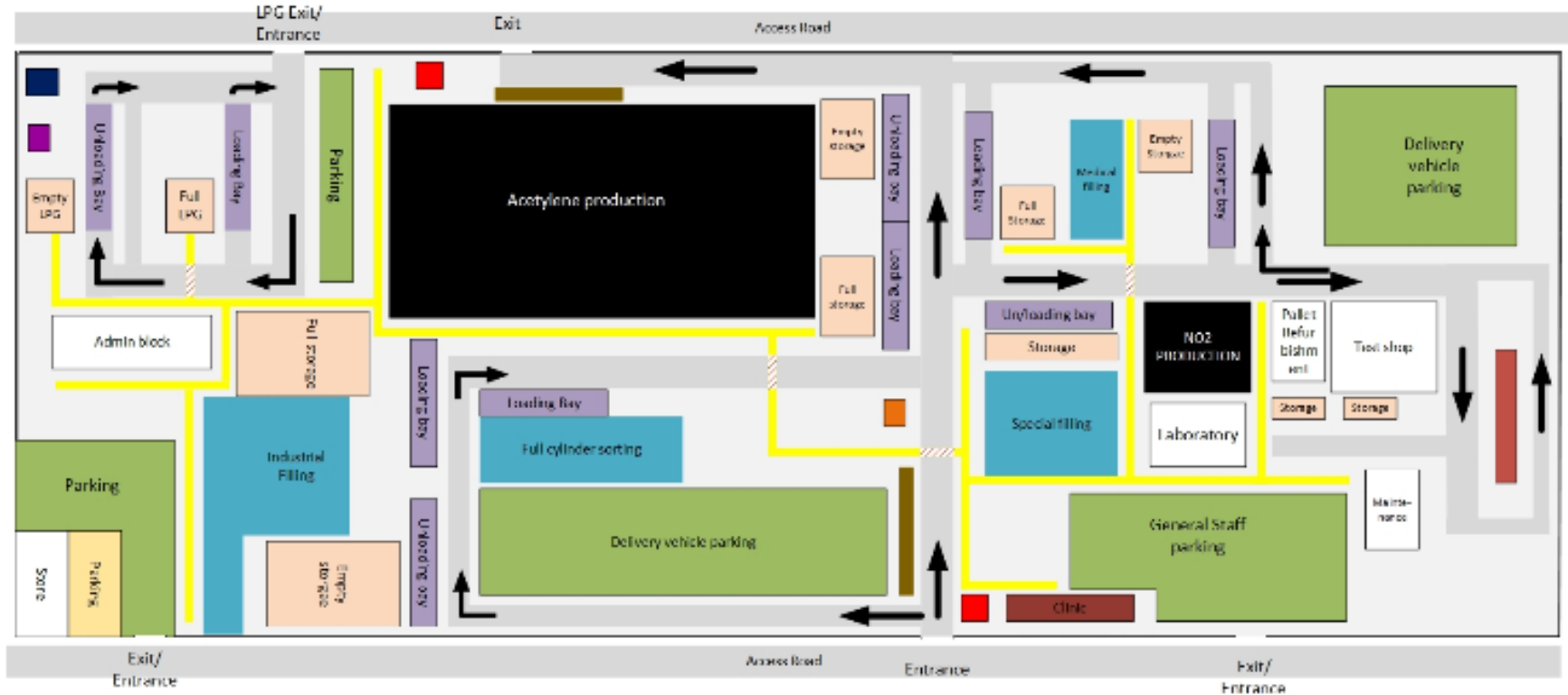
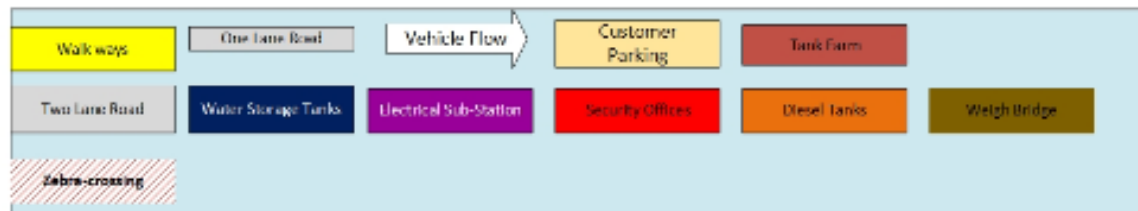
- Remember that you could have had Alternatives for only certain areas, aspects or material handling systems
- Intention of the MASTER PLAN (MFP) = the BIG picture
 - The overall 'picture'...
 - must be **visual** and must be **clear**
 - 2D or 3D (if it is an important factor)
 - (also called) Total Site/Plant layout or Plot Plan
 - Must indicate setting/context of public roads/waterways
- Factors to consider include:
 - Aesthetics (attractiveness, within a setting)
 - Expansion (10 yrs into future)
 - Flow patterns (outside integration to inside)
 - Sustainable energy usage (Smart/Green)

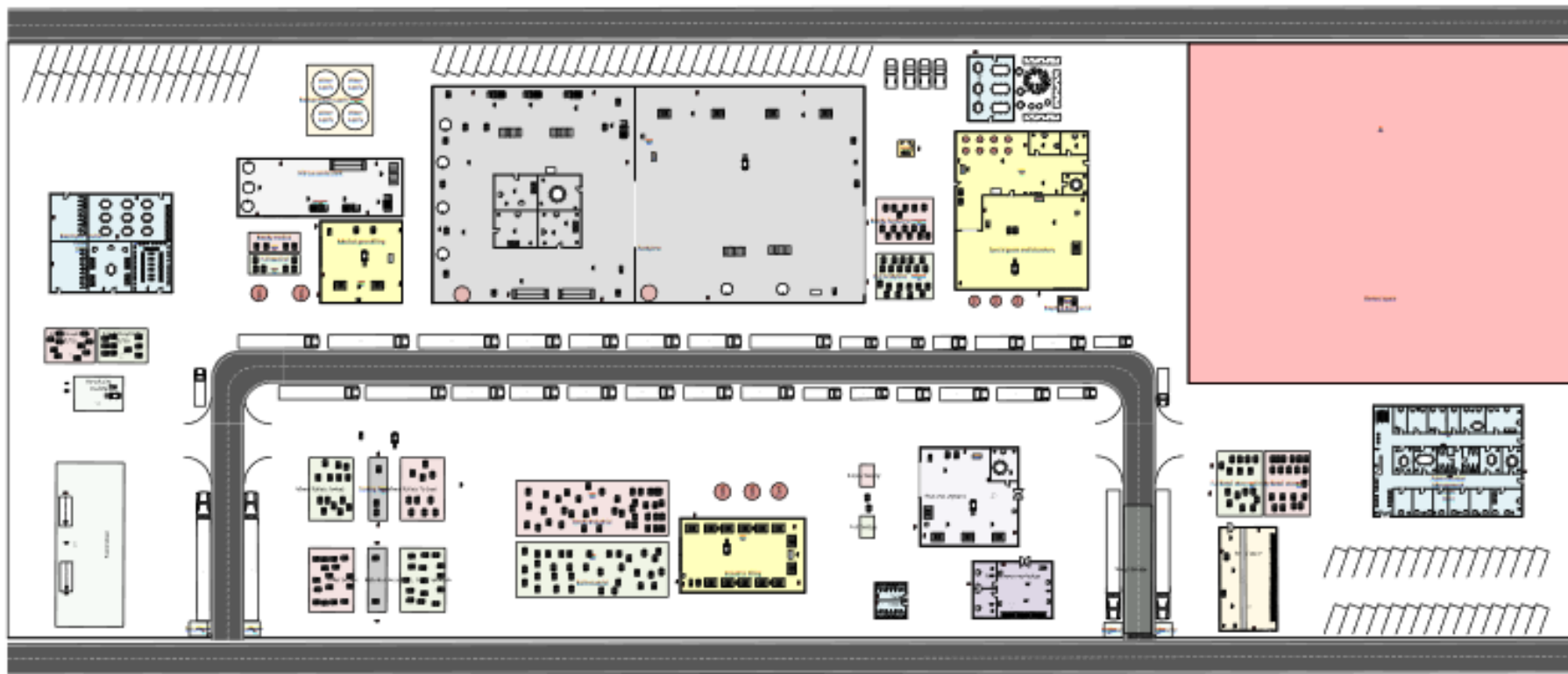
The Master Facility Layout *(Ch 12)*

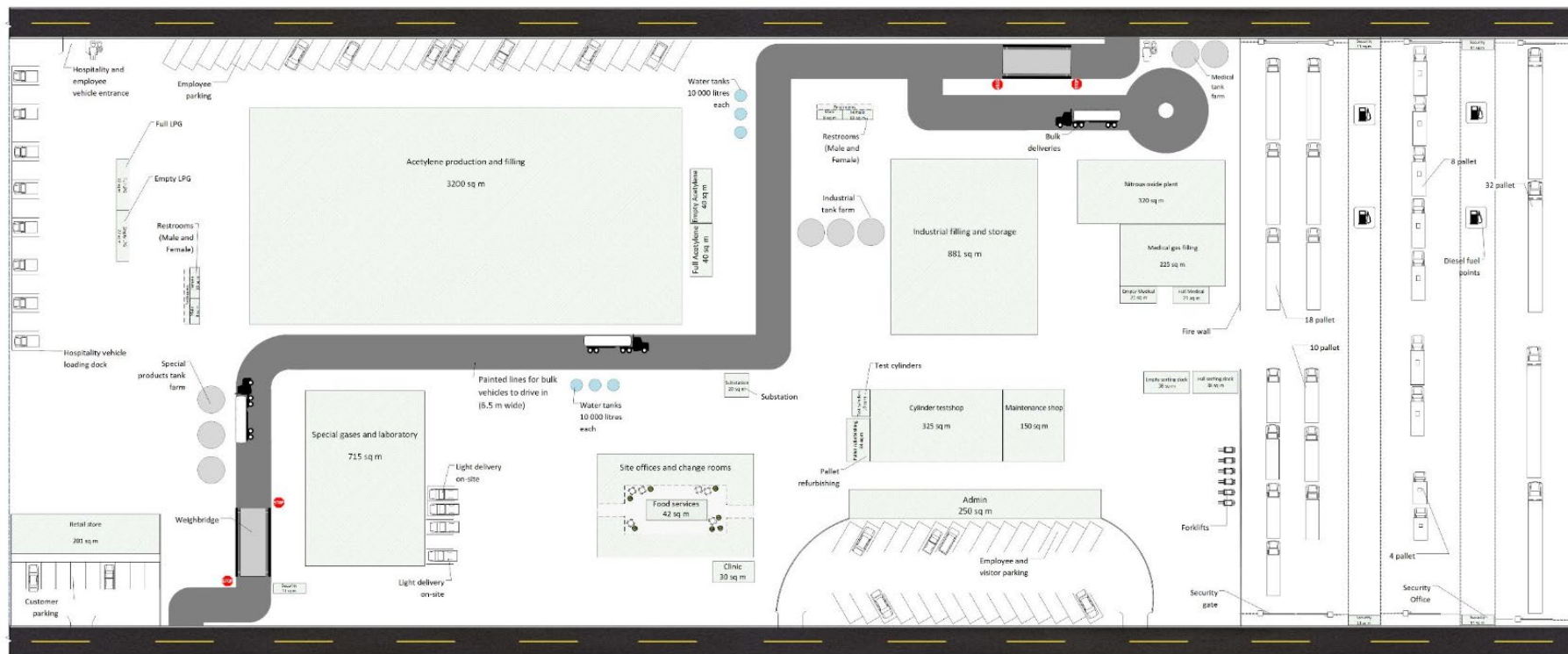
- Remember the logic you followed...
 - Start with setting/context of plot/land
 - Indicate outside and inside roads where applicable
 - Receiving and Shipping points
 - Warehouses (if outside)
 - Outside parking, security, offices (if outside manufacturing)
 - Outside walls
 - Production department blocks based on based on Space relationships charts and alternatives
 - Aisles
- Audit the layout....by following material flow
 - Is everything there?
 - Is flow smooth?
- Does the MFP align with block layouts?

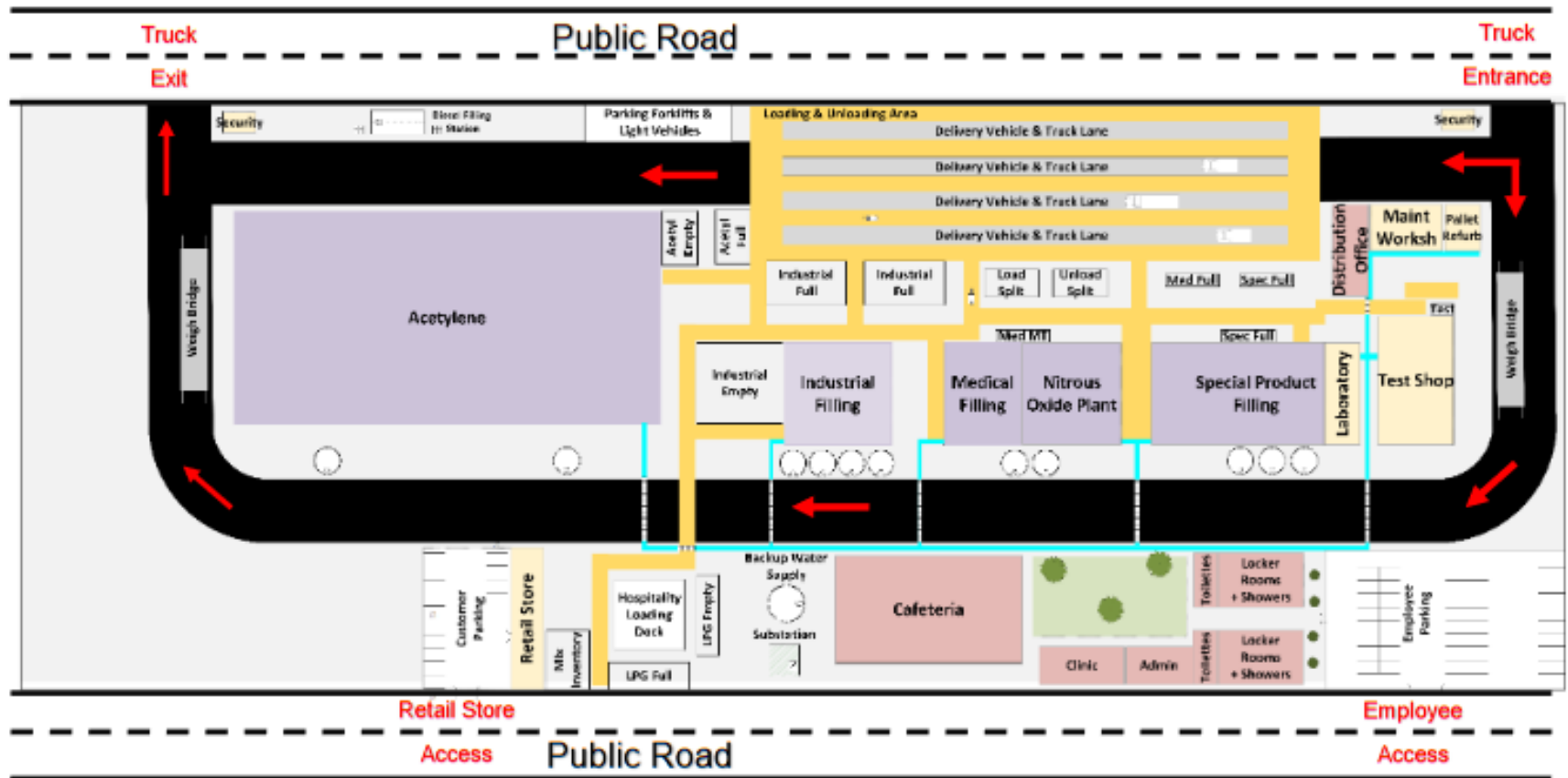
Some examples (Ch 12)











The Business case

- Define a 'case' for the project = to get the facility there so operations can start
- The start AND ending of the project should have been defined at the start
- Assume that it does not include stocking and starting production (unless specified in the real world)
- Include 3 things: BUDGET, SCHEDULE and RISKS (for purposes of this subject)
- Budget = includes the CAPITAL expenses for the building of the facility (not the operating expenses)
- Schedule = implementation plan of the building project and purchasing of the equipment (highlevel)
- Risks = things that can go wrong, and mitigation options

Selling the plan!

- You have WORDS and PICTURES to communicate with
 - Words in a Report (refer to guidance) somebody can READ or HEARD in a presentation (if required)
- Often ONLY the MFP is required (but must be able to **justify** and **support** calculations and decisions)
- For this module – challenge is INTERVIEW = questions asked on what/how/why to authenticate and evaluate practical insight
- Most NB – show your logic, logically

